**QP CODE : 26900159**

**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

**MGU-UGP (Honours) REGULAR EXAMINATION October 2026**

**Fifth SEMESTER**

**Discipline Specific Core Course (DSC) - MG5DSCCFS300 – DATABASE MANAGEMENT SYSTEM AND SECURITY**

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes Maximum Marks: 50

**Part A**

Very Short Answer Questions

Answer any **10** questions

Each question carries **1** marks

1.The concept of physical data independence - physical data independence is the capacity to change the internal schema without having to change the conceptual schema

2.The concept of subclass and super classes in EER model -

Superclass : A superclass is an entity type that contains common attributes shared by one or more related entity types.

Subclass: A subclass is an entity type that inherits the attributes and relationships of a superclass and may have its own additional attributes. Write suitable examples for both classes

3.Differentiate between a primary key and a candidate key

Primary key :

- A Primary key is used to uniquely identify each record in a table.
- Only one primary key can exist in a table.
- Cannot contain NULL values.

Candidate Key :

- An attribute or set of attributes that can uniquely identify each record.
- A table can have multiple candidate keys.
- Cannot contain NULL values.

4.Different levels of data abstraction : - The three levels of data abstraction in DBMS are:

- Physical level: Describes how data is stored in the database.
- Logical level: Describes what data is stored and the relationships among the data.
- View level: Describes only the part of the database that is relevant to a particular user.

5. The various file organization techniques used in DBMS are:

1. Heap (Unordered) File Organization
2. Sequential File Organization
3. Hash File Organization
4. Indexed File Organization
5. Clustered File Organization

6. Purpose of an Index in DBMS: An index is used to speed up data retrieval from a database table by allowing records to be located quickly without scanning the entire table.

7. The meaning of the symbol | in TRC - In Tuple Relational Calculus (TRC), the symbol | means "such that." It separates the tuple variable from the condition it must satisfy.

8. The meaning of the symbol ∧ in DRC - In Domain Relational Calculus (DRC), the symbol ∧ means logical AND (conjunction). It combines two conditions that must both be true.

9. One security measure used to prevent unauthorized database access : -

Password authentication is one security measure used to prevent unauthorized database access. Or User privileges and permissions Or Database auditing and monitoring or Data encryption etc . Write any and explain it briefly

10. Classify the security levels used in a multilevel security system - The four security levels used in a multilevel security system are:

1. Top Secret
2. Secret
3. Confidential
4. Unclassified

11. Employ a log record to cover an update made by a concurrent transaction - A log record used to record an update made by a concurrent transaction contains the transaction ID, data item, old value, and new value. This supports database recovery using log-based recovery.

12. Identify the storage types used for maintaining recovery information - The storage types used for maintaining recovery information in DBMS are:

1. Volatile storage
2. Non-volatile storage
3. Stable storage

## Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

13. Difference Between Entity Integrity and Referential Integrity Constraints

1. Entity Integrity Constraint

Entity integrity ensures that the primary key of a table cannot contain NULL values. It guarantees that every record in a table can be uniquely identified.

Example: In a STUDENT table, Student_ID is the primary key and cannot be NULL.

2. Referential Integrity Constraint

Referential integrity ensures that a foreign key value must match an existing primary key value in the referenced table or be NULL, if NULL is permitted. It maintains consistency between related tables.

Example: In an ENROLLMENT table, Student_ID is a foreign key referencing Student_ID in the STUDENT table. The foreign key cannot refer to a student who does not exist.

14. The symbols used in ER Model to distinguish between simple, composite, multivalued , derived and key attributes.

1. Simple attribute: this attribute cannot split sub part. It represented by a single oval.

2. Composite attribute: This attribute can be divided into sub parts, which reperesent more basics attributes with independent meaning . An oval connected to smaller ovals.

3. Multivalued attribute: An attribute that can have multiple values for a single entity. Represented by a double oval.

4. Derived attribute: An attribute whose value is calculated from another attribute. Example: Age derived from Date of Birth. Represented by a dashed oval.

5. Key attribute: An attribute that uniquely identifies each entity in an entity set. Example: Student_ID. The attribute name is underlined.

15. The applications of sequential file organization with suitable examples -

Sequential file organization is a method of storing records in a file in a specific sequence, usually based on a key field.

## Applications with Examples

1. Payroll Processing: Employee records are stored in employee ID order to calculate salaries. Example: Processing monthly salaries of all employees.

2. Banking Systems: Account or transaction records can be processed sequentially. Example: Generating daily transaction reports.

3. Billing Systems: Customer records are processed in sequence to prepare bills. Example: Generating monthly electricity bills. Etc

16. Different DML statements in SQL - DML consists of SQL statements used to retrieve, insert, modify, and delete data in a database table. DML statements are :-

1.Insert , 2. Update 3.Delete explain each with suitable example and also writs its purpose.

17. Classify valid and invalid foreign-key values in a student relation and show their effect on referential integrity :- Referential integrity ensures that a foreign-key value in one table matches an existing primary-key value in the referenced table, or is NULL if permitted. Example for Creating following tables DEPARTMENT and STUDENT.

1. DEPARTMENT Table (Parent Table)

| Dept_ID (Primary Key) | Dept_Name |
|---|---|
| 10 | Computer Science |
| 20 | Commerce |
| 30 | Mathematics |

2. STUDENT Table (Child Table)

| Student_ID (Primary Key) | Name | Dept_ID (Foreign Key) | Validity |
|---|---|---|---|
| 101 | Anu | 10 | Valid |
| 102 | Ravi | 20 | Valid |
| 103 | Meera | 40 | Invalid |
| 104 | Arun | 30 | Valid |
| 105 | Neha | NULL | Valid, if NULL is permitted |

Classification of Foreign-Key Values

1. Valid foreign-key values: Values 10, 20, and 30 are valid because they exist in the DEPARTMENT table.

2. Invalid foreign-key values: Value 40 is invalid because it does not exist in the parent table. Inserting this value violates referential integrity.
3. NULL foreign-key value: NULL is allowed if the foreign-key column permits NULL values and the relationship is optional.

Effect on Referential Integrity

- Valid foreign-key values maintain consistency between the tables.
- Invalid foreign-key values are rejected by the DBMS when referential integrity constraints are enforced.

18 .Use GRANT options to construct a privilege-propagation scenario involving users A1,A2 and A3 : - The GRANT statement is used to provide privileges to users. The WITH GRANT OPTION allows a user to pass the granted privileges to other users. .

Eg . Scenario

Assume user A1 owns a table named STUDENT and wants to grant privileges to A2 and A3.

**Step 1**: A1 grants privileges to A2

```
GRANT SELECT, INSERT ON STUDENT TO A2 WITH GRANT OPTION;
```

A1 grants SELECT and INSERT privileges to A2. Since WITH GRANT OPTION is specified, A2 can grant these privileges to other users.

**Step 2**: A2 grants SELECT privilege to A3

```
GRANT SELECT ON STUDENT TO A3;
```

A2 grants the SELECT privilege to A3. A3 can retrieve records from the STUDENT table but cannot grant the privilege to another user because WITH GRANT OPTION was not specified.

Part **C** Essay Questions

Answer any **2** questions

Each question carries **10** marks

19. The major limitations of traditional file processing systems - A traditional file processing system is a system in which data is stored and managed using separate files for different applications. Each application program maintains its own files and procedures to access and manipulate the data.

List Various Limitations - 1. Data Redundancy 2. Data Inconsistency 3. Difficulty in Accessing Data 4. Integrity Problems etc and explain each In detail

20. Differentiate aggregate functions and sort functions.

- Aggregate functions - perform calculations on a set of rows and return a single result. Various aggregate functions are – SUM() , AVG(), COUNT () ,MIN() ,MAX() explain each with suitable example

- Sorting is performed using the ORDER BY clause in SQL. It arranges records in ascending or descending order.

1. ASC ()

Eg. SELECT * FROM STUDENT ORDER BY Name ASC

2. DESC ()

SELECT * FROM STUDENT ORDER BY Mark DESC;

Illustrate the use of aggregate functions and ORDER BY with suitable SQL statements

21. Implement a checkpoint in a given transaction log and illustrate how it reduces recovery work.

A checkpoint is a recovery mechanism in DBMS that records a known point in the transaction log so that the system can reduce the amount of log information it needs to examine after a system crash.

Without checkpoints, the recovery manager may need to scan a large portion of the log from the beginning. With checkpoints, it can usually start from a more recent recovery point, while still examining any earlier records required by the recovery protocol.

Write example of Transaction Log and write implementation of Checkpoint record into transaction log . And also write how it reduces recovery work.